

TIME SERIES ANALYSIS & CLASSIFICATION

T. MEDKOUR & H. BRAIRI

NATIONAL SCHOOL OF ARTIFICIAL INTELLIGENCE

Chapter VII: Model Diagnostics

1. Model Diagnostics

MODEL DIAGNOSTICS

Topics Covered So Far

Until now, we have discussed the following topics:

- ▶ Model specification (model selection): Choosing values of p, d, q most consistent with observed data.
- ▶ Model fitting (parameter estimation): Estimating parameters in the $\text{ARIMA}(p, d, q)$ class.

Model Diagnostics

- ▶ In this chapter, we focus on model diagnostics, which involves checking the fit of an $ARIMA(p, d, q)$ model.

- ▶ Previously we introduced this process for deterministic trend models:

$$Y_t = \mu_t + X_t, \quad \text{where } E(X_t) = 0.$$

- ▶ We will see that similar diagnostic techniques apply to $ARIMA$ models.

Residual Analysis

Terminology: Residuals represent unexplained variation in Y_t after model fitting.

$$\text{Residual}_t = \text{Observed } Y_t - \text{Fitted } Y_t.$$

► $AR(p)$ Model:

$$Y_t = \theta_0 + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + e_t,$$

where $\theta_0 = \mu(1 - \sum_{j=1}^p \phi_j)$ is the intercept term.

Residual Calculation

- ▶ Residual at time t is computed as:

$$\hat{e}_t = Y_t - (\hat{\theta}_0 + \hat{\phi}_1 Y_{t-1} + \cdots + \hat{\phi}_p Y_{t-p}).$$

- ▶ Residuals serve as proxies for white noise terms e_t , aiding in model assessment.

Normality and Independence Diagnostics

- ▶ Histograms and QQ plots help visually assess normality.
- ▶ Time series plots of residuals detect patterns violating independence.
- ▶ Formal tests:
 - Shapiro-Wilk test: Tests normality.
 - Runs test: Tests independence.

Residual ACF and Ljung-Box Test

Recall: For white noise, residual ACF satisfies:

$$r_k \sim \mathcal{AN}(0, 1/n), \quad \text{for large } n.$$

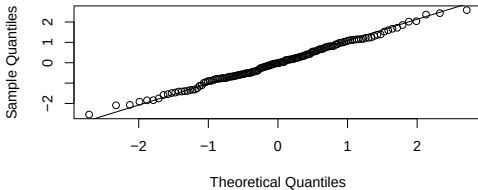
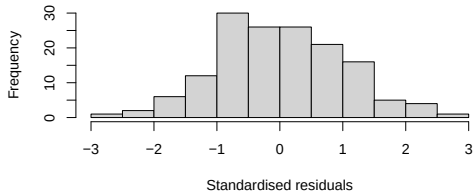
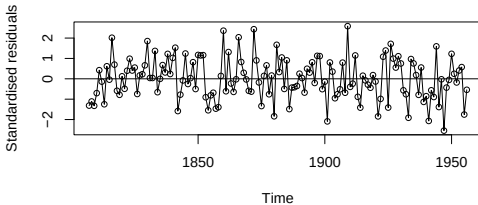
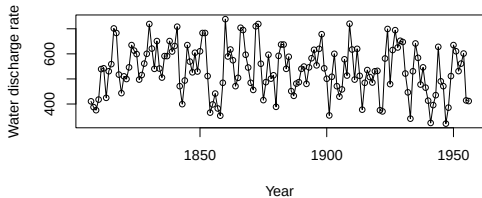
- Ljung-Box Test: Evaluates overall residual autocorrelation:

$$Q^* = n(n+2) \sum_{k=1}^K \frac{\hat{r}_k^2}{n-k}.$$

Tests:

- H_0 : The ARMA(p, q) model is appropriate.
- H_1 : The ARMA(p, q) model is not appropriate.

Example



Shapiro-Wilk Normality Test

```
> shapiro.test(rstandard(gota.ma1.fit))
```

Shapiro-Wilk normality test

```
data:  rstandard(gota.ma1.fit)
```

```
W = 0.9951, p-value = 0.8975
```

Runs Test

```
> runs(rstandard(gota.ma1.fit))
```

```
$pvalue
```

```
[1] 0.29
```

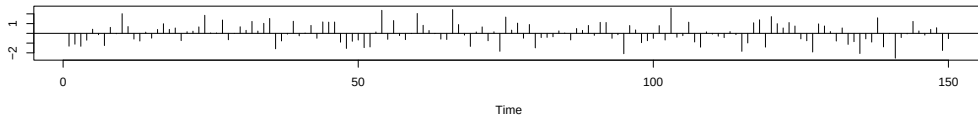
```
$observed.runs
```

```
[1] 69
```

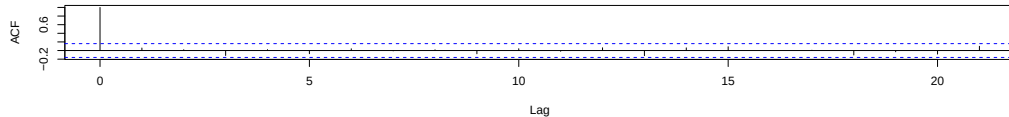
```
$expected.runs
```

```
[1] 75.94667
```

Standardized Residuals



ACF of Residuals



p values for Ljung-Box statistic

